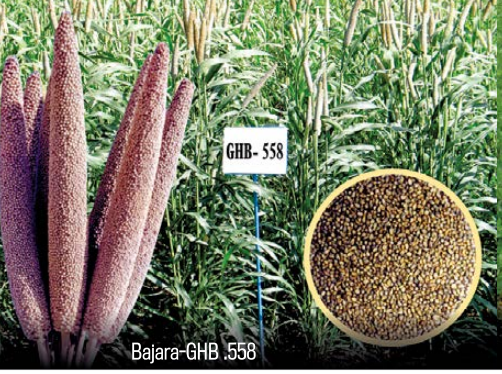
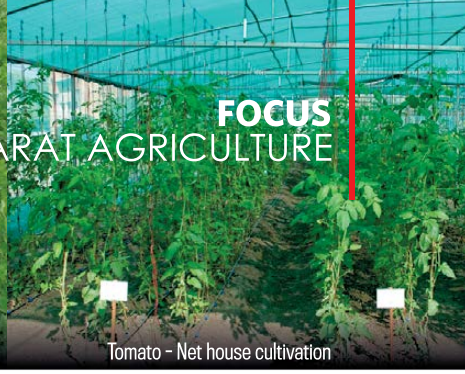


Bajara-GHB .558


Groundnut - GG.23 (Sorath Kiran)


Tomato - Net house cultivation

THE PIONEERING ROLE OF GUJARAT'S STATE AGRICULTURAL UNIVERSITIES IN AGRICULTURAL INNOVATION

A LEGACY OF AGRICULTURAL EXCELLENCE

The four State Agricultural Universities (SAUs) of Gujarat—Anand, Junagadh, Navsari, and Sardar Krushinagar—were established in 2004 by carving them out from Gujarat Agricultural University (GAU). Since their inception, these universities have become instrumental in transforming Gujarat's agricultural landscape through education, research, and technological advancements, thereby contributing significantly to the state's economic growth and agricultural productivity.

EDUCATIONAL INFRASTRUCTURE AND OUTREACH

These universities currently operate 30 degree colleges, 23 polytechnic colleges with a 4000 intake capacity annually, and 4 postgraduate institutes. They also manage 114 research stations and 51 extension education centers spread across the state, enabling them to impact a wide range of agricultural practices. Through these institutions, Gujarat is nurturing the next generation of agricultural experts, while also providing farmers with cutting-edge knowledge and techniques to improve their productivity.

GROUNDBREAKING RESEARCH AND DEVELOPMENT

The SAUs have developed over 350 high-yielding, disease-resistant varieties of major crops in Gujarat. Their research focuses on bio-fortified varieties, climate-resilient crops, smart/precision farming, and sustainable agricultural practices. Notable achievements include the world's first hybrid cotton, the first public sector Bt-cotton, Rayan as a rootstock for sapota, Jaggery from sapota, and innovations like precision farming, organic farming, and protected cultivation. The universities have also made strides in developing technologies like closed sub-surface drainage, brackish water aquaculture, and biofertilizers.

The productivity of castor, for example, has risen dramatically from 368 kg/ha in 1969 to 2100 kg/ha in 2021-22, with Gujarat now accounting for 71.04% of India's total castor production, which constitutes 81% of global castor output. Similarly, wheat research has led to a significant increase in yield, with productivity rising from 1970 kg/ha in 1989-90 to 3200 kg/ha in 2022-23. These contributions have made Gujarat a leader in wheat production, with varieties from the SAUs being popular not only in the state but also in neighboring areas like Madhya Pradesh, Rajasthan, and Central India.

IMPACT ON KEY CROPS

One of the most significant contributions is the cumin variety GC 4, released in 2005-06, which now covers more than 80% of Gujarat's cumin-growing area. Since its release, GC 4 has contributed 39.65 lakh tons to India's cumin production, generating an estimated export value of Rs 25,480 crore. The total monetization due to GC 4 in Central Gujarat is Rs 1893 crore. Additionally, the universities have played a crucial role in increasing pulse production, with productivity growing from 543 kg/ha in 2004 to 1471 kg/ha in 2021-22, contributing to India's self-sufficiency in pulses.

SEED DEVELOPMENT AND DISTRIBUTION

In seed production, the universities have developed a range of high-quality varieties, including the 'Anubhav Seed,' 'Gir Savaj Seed,' 'Navroji Seed,' and 'Banas Seed.' They have managed to produce large quantities of foundation, certified, and truthful seeds, ensuring that farmers have access to the best possible planting material.

LEADERSHIP IN TISSUE CULTURE AND BIOTECHNOLOGY

Gujarat's SAUs have also led the way in tissue culture, particularly in Date Palm cultivation, which has been commercialized to generate significant revenue. The universities have developed 11 micropropagation protocols and six regeneration protocols for various crops, contributing to the advancement of agricultural biotechnology.

A FUTURE OF SUSTAINABLE AGRICULTURAL GROWTH

Gujarat's State Agricultural Universities have significantly impacted agriculture not only in Gujarat but also nationally and globally. With over 350 crop varieties developed, breakthrough technologies, and a strong educational infrastructure, they continue to lead the way in sustainable agricultural practices. Their relentless innovation ensures that Gujarat remains at the forefront of agricultural development, fostering food security, economic prosperity, and environmental sustainability.


Agri photo voltic at Junagadh